UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 010 · Problems on maxima and minima for a function of two variables · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a

Contents

1. Figure G1.1 · Level curves of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$
2. Figure G1.2 · Level curves of $x^3 - 12x + y^3 + 3y^2 - 9y$
3. Figure G2.1 · Level curves of $x^4 + y^4 - x^2 - y^2 + 1$
4. Figure G3.1 · Temperature $400xy(1 - x^2 - y^2)$ over the unit disk
5. Figure G3.2 · Metal used by an open box of 10 m³ with a square base

The deck has no graphs, so every figure is added for clarity and drawn from the deck's and the papers' own functions. Contour plots: thin grey lines are level curves; a filled vermilion triangle marks a maximum, a filled blue circle a minimum and a green cross a saddle point, each labelled directly.

1 The method, and the ESA cubic functions

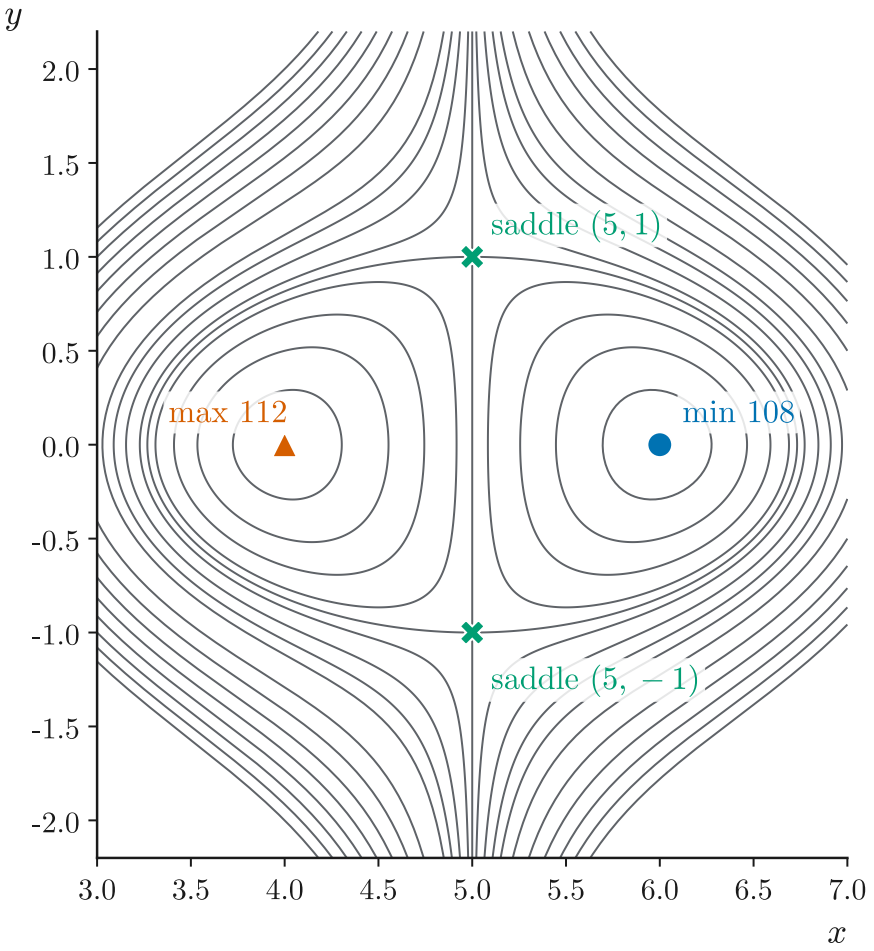


Figure G1.1. Level curves of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. ESA ★★★

ADDED FOR CLARITY S010·8 · ESA Dec-23 1c · ESA Jan-24 s1 1a

How to read it. Grey lines are level curves $f =$ constant. Around the maximum $(4, 0)$ and the minimum $(6, 0)$ they close into small loops; at the saddle points $(5, \pm 1)$ two level curves cross, the sign of the picture of a saddle.

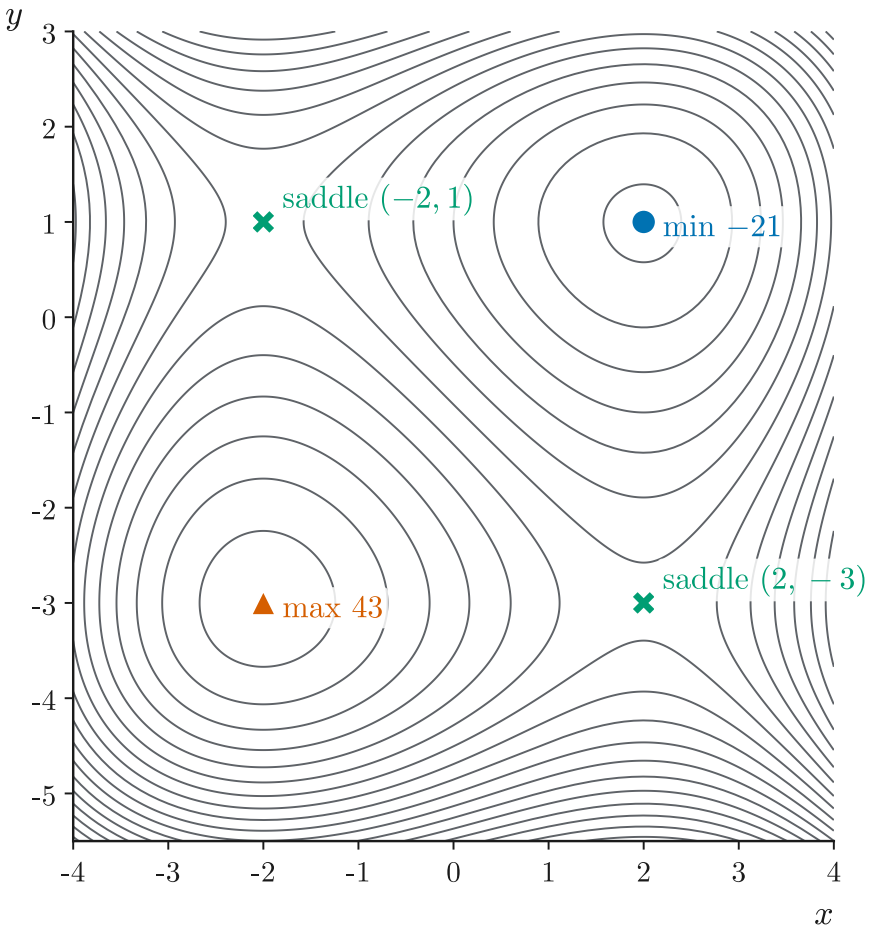


Figure G1.2. Level curves of $x^3 - 12x + y^3 + 3y^2 - 9y$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 1a · CW U1 P29

How to read it. The four stationary points sit at the corners of a rectangle, $x = \pm 2$, $y = 1, -3$. Closed loops surround the maximum $(-2, -3)$ and the minimum $(2, 1)$; curves cross at the saddles $(2, -3)$ and $(-2, 1)$.

2 Functions with many stationary points

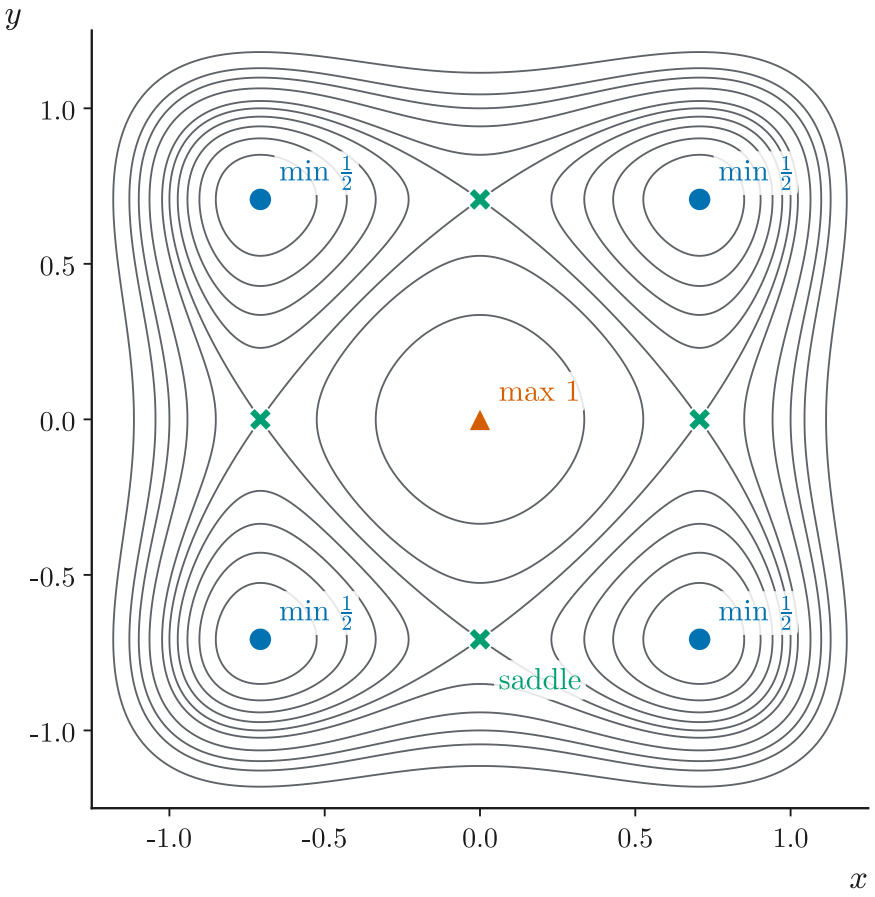


Figure G2.1. Level curves of $x^4 + y^4 - x^2 - y^2 + 1$. ESA ★★

ADDED FOR CLARITY S010·11 · S010·12

How to read it. Nine stationary points on a $\mathbf{3 \times 3}$ grid: a maximum at the centre, minima at the four corners $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$ and saddles at the four edge midpoints on the axes.

3 Applied problems: eliminate one variable first

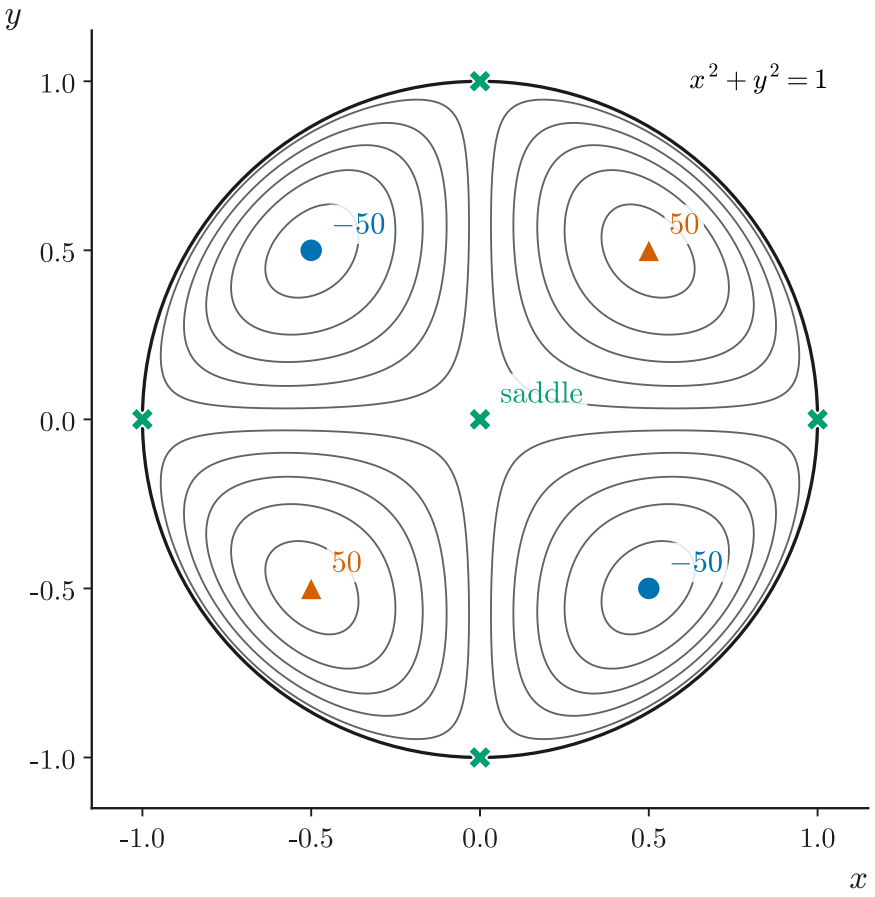


Figure G3.1. Temperature $400xy(1 - x^2 - y^2)$ over the unit disk. ESA ★★

ADDED FOR CLARITY S010·3 · S010·4

How to read it. Inside the unit circle (the projection of the sphere) the temperature is positive in the first and third quadrants and negative in the others. It peaks at 50 at $(\tfrac{1}{2}, \tfrac{1}{2})$ and $(-\tfrac{1}{2}, -\tfrac{1}{2})$ and is lowest, -50, at the other two. The origin and $(\pm 1, 0)$, $(0, \pm 1)$, where $T = 0$, are saddles.

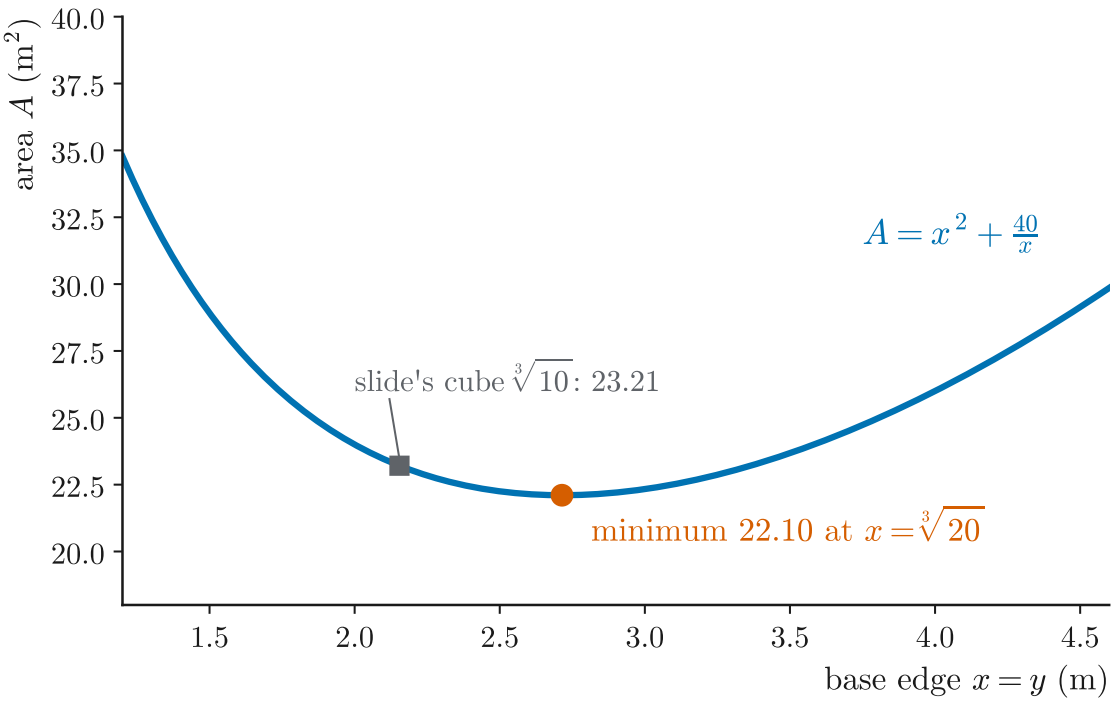


Figure G3.2. Metal used by an open box of 10 m³ with a square base. ESA ★

ADDED FOR CLARITY S010·6 · CW U1 P34

How to read it. Along $x = y$ the area is $A = x^2 + \frac{40}{x}$ (blue). Its minimum, **22.10** m², is at $x = \sqrt[3]{20} \approx 2.71$ m. The slide's cube, $x = \sqrt[3]{10} \approx 2.15$ m, sits higher on the curve (**23.21** m²).